

Introduction

Carbon trading is a common market-based tool to reduce Green House Gas (GHG) emissions and tackle the climate crisis. Uptake of carbon trading has expanded greatly, and they now cover over 23% of global GHG emissions (International Carbon Action, 2025). Yet questions remain over their effectiveness and cost-efficiency compared to alternative policy measures (Metcalf, 2023). This essay compares the European Union Emissions Trading Scheme (EU ETS) with California's Cap-and-Trade (C&T) scheme, evaluates their effectiveness, and discusses if alternative mechanisms are more suitable.

Emissions trading gives regulators direct control over total emissions while theoretically minimising the total cost of emissions abatement. Imposing tradeable permits enables the market to find the most cost-efficient method of abatement. Regulators have flexibility over the initial allocation of permits, which can be allocated via auction using an 'emitter pays' principle to generate an income stream for government.

The EU ETS is the largest scheme of this type in the world, and the only supranational example. Participating countries include the 27 EU states plus Norway, Iceland, and Liechtenstein. Launched on January 1st, 2005, the scheme is the oldest still in operation and accounts for about 40% of the EU's emissions (International Carbon Action, 2025).

California's C&T program is an example of emissions trading on a regional level. The scheme has been operational since 2012 and covers 76% of the State's GHG emissions (International Carbon Action, 2025).

Since becoming operational, emissions within the EU ETS and California C&T have significantly reduced. Energy generation and industrial emissions covered under the EU ETS reduced by 37.3% between the scheme's launch in 2005 and 2021 (European Commission, 2023). Emission reductions from installations under California's C&T scheme aren't published directly, but sectoral data shows a total reduction in electricity generation plus industrial emissions of 26% between 2013 and 2023 (California Air Resources Board, 2025a).

Nevertheless, these reductions may be attributed to a variety of external factors including other climate policy measures. After evaluating the schemes, alternatives including regulation and carbon tax are discussed, and the importance of cost-efficiency is highlighted.

Comparison

The carbon-trading schemes of California and the European Union are similar at a basic level. Permits are granted or auctioned to emitters covered by the scheme, and these firms can then trade them as they see fit. Both schemes cover a variety of GHG gasses on a CO₂ equivalent (CO₂e) basis. The total volume of permits granted decreases annually in line with climate targets. Differences in configuration of the systems may however contribute to different outcomes. The main points of comparison are now expanded on. Unless otherwise stated, information is extracted from the International Carbon Action Partnership Status Report, (2025).

Targets

- **EU ETS:** Net GHG emissions reduction of 55% from 1990 levels by 2030. Linear reduction in emissions cap.

- **California C&T:** Less ambitious than the EU, California is targeting a net GHG emissions reduction of 40% from 1990 levels by 2030. Roughly linear reduction in emissions cap.

Scope

- **EU ETS:** Expanded in 2024, the scheme now covers emissions¹ from energy generation, manufacturing, internal aviation², and maritime transport. The scheme includes over 8,500 installations, 379 aircraft operators, and 2251 shipping companies.
- **California C&T:** Broader in scope than the EU ETS, this scheme covers sector emissions³ from mining, power, buildings, transport, industrial, agriculture, and forestry. Approximately 400 installations are included.

Allocation principle

- **EU ETS:** 57% auctioned, the remained is primarily free allocation with some held in a stability reserve. 10% of free allocation goes to lower-income member states. Free allocation based on performance benchmarks including technology progress, risk of leakage⁴, as well as provisions for fast-growing companies and new market entrants. The benchmarking is notably nuanced (European Commission, 2024).
- **California C&T:** 46% auctioned. Similar to the EU, free allocation primarily for utilities to benefit ratepayers, and energy intensive industrial facilities to assistance transition, minimise leakage, and reward early movers (California Air Resources Board, 2025c).

Price Stability

- **EU ETS:** A Market Stability Reserve addresses a growing allowance surplus and protects against demand shocks by regulating the volume of new permits. There are also established future and options markets to promote hedging and smooth shocks.
- **California C&T:** An annually increasing auction price floor supports permit prices. Similarly to the EU, an Allowance Price Containment Reserve exists to provide extra permits if necessary, although there has been no such occasion to date.

Revenue Reinvestment

- **EU ETS:** The 2023 breakdown of fund usage by member states included project support for energy (43%), transport (23%), and Social and Just Transition (12%). A share of ETS revenues fund EU Innovation, Modernisation, and Social Climate funds.
- **California C&T:** Most auction revenues go to a ringfenced California Climate Investments fund (California Air Resources Board, 2025b). 73% of this has been targeted towards low-income and disadvantaged communities.

Governance

- **EU ETS:** The EU's largely unique Carbon Border Adjustment Mechanism (CBAM) accounts for carbon-intensive goods that are imported to the EU. The mechanism is used to reduce carbon leakage by ensuring imported goods under certain high-risk sectors is accounted for.

¹ On a CO₂e basis: CO₂, N₂O, HFCs, PFCs.

² Coverage is restricted to flights within the European Economic Area, to the UK and Switzerland.

³ On a CO₂e basis: CO₂, CH₄, N₂O, SF₆, HFCs, PFCs, NF₃, other fluorinated GHGs.

⁴ In competitive markets, energy intensive industry often cannot pass the high-cost of abatement through to buyers and are instead incentivised to move production outside the carbon-trading region, i.e. the risk of carbon leakage is high.

- **California C&T:** The scheme has been formally linked with Québec's Cap and Trade scheme since 2014. Unlike the EU ETS, California allows offsets and credits up to 4% of an installation's compliance obligation by trading with Québec.

Evaluation

Laing et al., (2013) reviewed the existing body of studies evaluating the first eight years of the EU ETS. They concluded that studies have in general found that the EU ETS has driven a reduction in emissions, although over-allocation has reduced this.

Difference-in-difference studies have been used by multiple studies to establish a causal link between the EU ETS and a variety of economic indicators. Marin et al, (2018) used this approach and firm-level matching to find that the EU ETS did not negatively impact economic performance and resulted in improved labour productivity. Similarly, Jaraitė and Maria, (2016) provided causal evidence that the EU ETS has increased investments in Lithuania without reducing profitability.

Kramer and Lessmann, (2024) evaluated California's C&T scheme over the 2005-2019 period using a synthetic control model. They found emissions from the power sector were reduced significantly compared to the counterfactual, but this was somewhat offset by a smaller increase in industrial emissions. Holliman and Collins, (2023) concluded the Cap-and-Trade system enhanced social and environmental equity but did not reduce emissions in line with California's goals. Furthermore, they identified evidence of regressive taxation of transport.

The literature review above is extended here by an empirical analysis of more recent data. The years 2019-2023 are used to compare the EU ETS and California's Cap and Trade scheme.

Average reduction in emissions (CO₂e) under the EU ETS is summarised in table 1. Maritime transport was only recently incorporated into the EU ETS and is excluded due to insufficient data. Notably EU Industrial Production has been flat over the period of analysis, experiencing only 0.06% average annual growth (Eurostat, 2025b). This must be seen in the context of strong industrial emissions reduction and possibly indicates carbon leakage. This is evidenced by steady annual GDP growth of 1.36% (World Bank, 2025). Employment has also increased by 3.6% over the period, indicating an largely healthy European economy (Eurostat, 2025a). The share of electricity production from renewable sources has increased from 34% in 2019 to 45% in 2023 (Our World in Data, 2025). Although not providing causal evidence, the analysis indicates an association between the ever-tightening EU ETS caps and a reallocation of economic activity away from carbon-intensive activity.

Table 1 Data calculated from [European Commission data](#)

Sector	Average CO ₂ e Annual Reduction (2019-2023)
Electricity & Heat Production	10.0%
Industrial Production	5.42%
Aviation ⁵	(10.9%)

⁵ Treatment of UK and Swiss flights has changed during this period, as well as flights from the EU's nine outermost regions. Yearly figures are highly volatile as a result, and the average provides limited insight.

Table 2 compiles sectoral emissions data for California over the same period⁶. Unlike the EU, Californian Industrial production has seen steady growth of the period of study while simultaneously steadily reducing emissions (Analysis, 2025). The wider economy has grown in tandem, with average annual GDP growth of 5.8% and healthy employment (U.S. Bureau of Economic Analysis, 2025). The notable increase in mining emissions over the period highlights the high cost of abatement in this industry, if it is possible at all.

Surprisingly, transport emissions have reduced annually by an average of 3.8%, indicating a decoupling from economic activity. This may in part be the result of a \$5 Billion reinvestment of Cap and Trade auction proceeds in transport projects (California Air Resources Board, 2025d). However, as noted by Holliman and Collins, (2023), the cost of this shift may be unduly carried by California's lower income population. The high level of auction funds directed to disadvantaged and low-income communities may negate this regressive effect, as argued by Metcalf, (2023). The success of transport sector emissions abatement provides a roadmap to the EU ETS 2 scheme, due to launch in 2028.

Table 2 Data calculated from California Air Resources Board data

Sector	Average CO₂e Annual Reduction (2019-2023)
Power Production	2.5%
Industrial Production	2.9%
Mining	(11.9%)
Buildings	(2.7%)
Transport	3.8%
Agriculture	2.0%

California's renewable share of electricity generation has increased by a total of 4% between 2019 and 2023. The most notable area of investment has been in electricity storage projects, which have boomed from virtually zero in 2019 to 17GW in 2025 (California Energy Commission, 2025). Although emissions from power production have steadily declined over the period, they were dwarfed by reductions in the EU. The contrast is even more stark when emissions from buildings are included. The 2022 energy crisis in Europe is a significant factor here. A second reason may be the introduction of Phase 4 of the ETS in 2021. The latest phase has further tightened emissions caps, constraining the supply of permits and putting upward pressure on the equilibrium price. The diverging trend in Figure 1 illustrates this.

⁶ Total sectoral emissions are used as a proxy to direct emissions data from California's C&T system. A portion of the emissions tabulated therefore comes from smaller installations not participating in the C&T scheme.

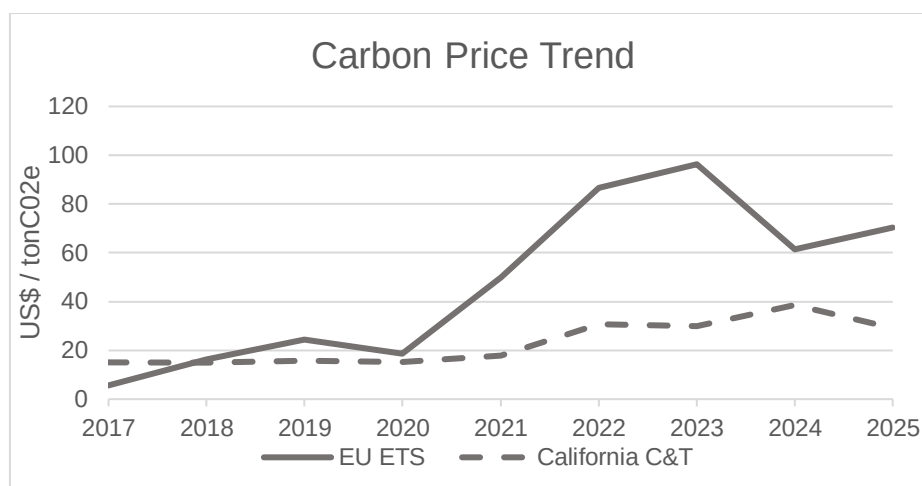


Figure 1 Data Source: <https://carbonpricingdashboard.worldbank.org/compliance/price>

Other Instruments

The main policy alternatives to carbon trading are carbon taxing or command and control regulations. Command and control regulations set an emissions ceiling for firms, and/or mandate the introduction of energy efficiency standards. Emissions ceilings are not typically preferred because although they do fix the level of abatement, they don't minimise the total cost achieving this. Efficiency standards are more common and are found in Both Europe and California. The EU Energy Efficiency Directive sets requirements at a high-level for Europe, delegating specific policies to member states (European Commission, 2025). Typical efficiency measures include home insulation requirements or fuel-efficiency of vehicles. Energy efficiency regulations are an appropriate tool to mandate the use of proven technologies. However, the direct approach does not incentivise free-market dynamics to deliver further efficiencies. This is why they are used in conjunction with a carbon tax or carbon trading.

Carbon Taxing is the main alternative to emissions trading, and regulates price as opposed to quantity. Taxing carbon and emissions trading are equivalent when the marginal cost and benefit of emissions abatement is known. In reality it is not known, and the efficiency of the two policies differ. Weitzman, 1974, found that price regulation is more efficient than quantity regulation when the marginal benefit curve is flat, and marginal cost is steep, as illustrated in figure 2 (Weitzman, 1974). Newell and Pizer, 2003, extend and apply this model to the problem of policy selection for greenhouse gas abatement. They find that taxing carbon is preferable to carbon trading (Newell & Pizer, 2003).

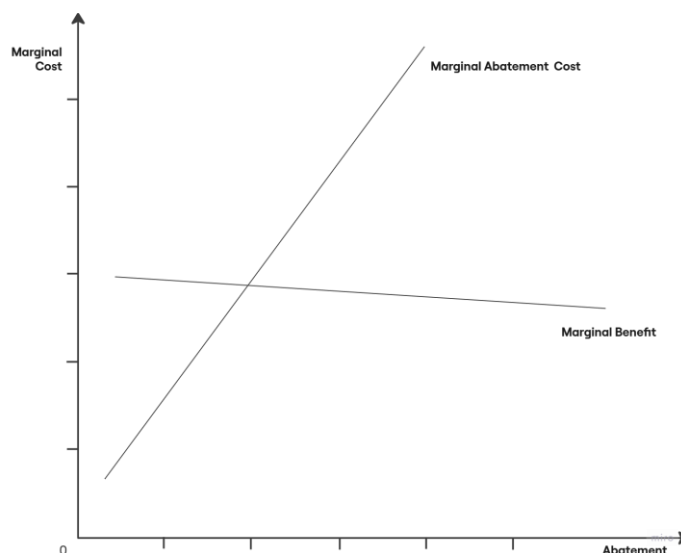


Figure 2: Marginal Cost and Benefit curves of GHG Emissions Abatement

The constraints of this theoretical model do not reflect the complexity and dependencies of policy making. For one, the cost of dismantling the EU ETS or California C&T systems to replace them with a carbon tax may immediately outweigh any near-term benefits. The urgency of abating GHG emissions quickly undermines such a drastic switch in policy. Such a change might be more feasible on the regional scale in California rather than at a supranational level in Europe. Another flaw of the model is the assumption that the marginal benefit of abatement is flat, i.e. a linear return on emissions abatement. The assumption is based on theoretical simulation rather than longitudinal empirical study. Considering the strategic complementarities of rapid emissions abatement within a wider policy strategy may realise a steeper concave marginal benefits curve (Coyle, 2022; Karkatsoulis et al., 2016). This finding is particularly relevant to California, a hotbed of innovation. Brink et al, 2016, also argue that tightening the emissions cap help improve the efficiency of emissions-trading (Brink et al., 2016).

Another issue with carbon tax is that the regulator doesn't initially know the level of tax that will prompt the desired level of abatement. Once a tax is set, firms will adjust their output so that the marginal cost of one further unit of output equals the marginal cost of production including tax. This minimises the overall cost of abatement, but the overall abatement level is unknown in advance. This Regulator's Dilemma extends the time required to achieve the desired level of abatement through trial and error of tax adjustments, undesirable in the context of an urgent climate crisis.

Of course, the two Cap and Trade systems are not without their flaws. Most pressing is the risk of carbon leakage as firms shift production outside the scope of the trading scheme. CBAM (essentially a tariff on imported goods) mitigates this risk in Europe, but it remains highly prevalent in California. Both schemes use free allocations of permits to support industries at risk, but the complex methodology may discourage new market entrants.

Finally, political considerations important. Taxing carbon is particularly visible and often unpalatable. The failed introduction of a carbon tax in France in 2009 is one example of this (France24 News, 2009). Owning the status quo is thus a weighty argument in favour of carbon trading.

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